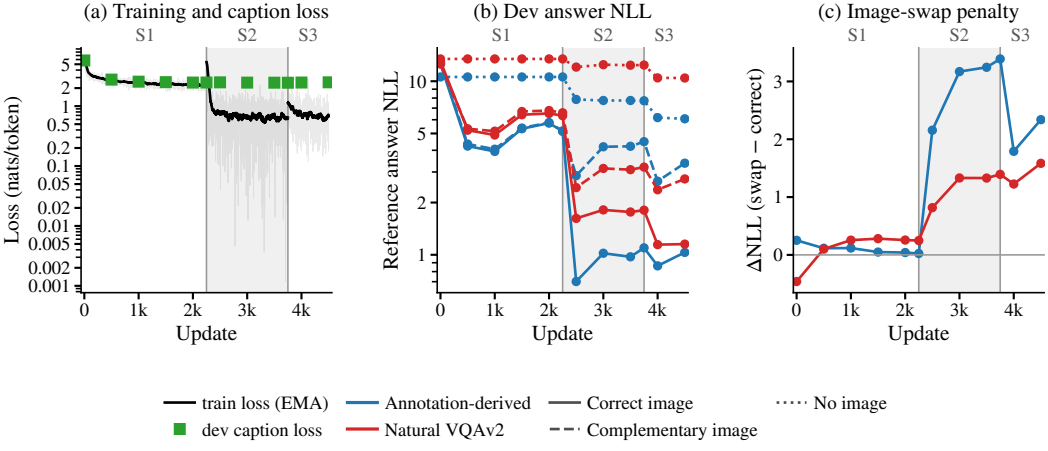
(a) Training and caption loss
(b) Dev answer NLL
(c) Image-swap penalty
S1
S2
S3
Loss (nats/token)
Reference answer NLL
ΔNLL (swap − correct)
Update
train loss (EMA)
dev caption loss
Annotation-derived
Natural VQAv2
Correct image
Complementary image
No image